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RESEARCH PAPER

Nanofluidic optical diffraction interferometry for detection and classification of individual nanoparticles in a nanochannel

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Abstract
Detection and classification of nanoparticles in flow at a single particle resolution are becoming extremely important with the progress in materials chemistry, life science, and nanotechnology. Interferometric light scattering has been widely used as a highly sensitive nanoparticle detection technique. However, its application to flow cytometric analyses is challenging, because precise fluidic control and complicated optical system are inevitable for accurate measurement of individual nanoparticles in flow. Here, we report the development of nanofluidic optical diffraction interferometry (NODI) which utilizes optical diffraction by a single nanochannel as a reference light for interferometric light scattering detection of nanoparticles in the nanochannel. Our method detected individual gold nanoparticles as small as 20 nm in diameter in flow by a simple optical system. Furthermore, by introducing a dual-wavelength measurement system, four types of gold and silver nanoparticles with 40–60 nm in diameter were discriminated based on their interferometric signals. Using a support vector machine (SVM) algorithm, classification of individual nanoparticles was achieved with over 70% accuracy. Our simple yet sensitive detection method will be widely used for the quantification and characterization of various nanoparticles.

Keywords Nanofluidics · Nanochannel · Diffraction · Light scattering · Interferometry

1 Introduction

Detection and identification of nanoparticles at a single particle level are of growing interest with the progress in chemical/biological applications of engineered nanoparticles and the increase in scientific and clinical interest for nanoscale chemical/biological targets. For example, synthesized or engineered nanoparticles such as plasmonic metal nanoparticles, lipid vesicles, and quantum dots have been used for drug delivery systems (Lombardo et al. 2019), in vivo therapy (Liu et al. 2019), and nanomedicine (Shi et al. 2017). From a clinical point of view, viruses and extracellular vesicles which are typically smaller than 100 nm in size are becoming major targets for diagnostics (Zhang et al. 2019; Corman et al. 2020; Kalluri and LeBleu 2020). Label-free, high-throughput detection methods are essential to quantify and evaluate such nanoparticles with a single particle resolution and statistically sufficient number of counts. Flow cytometric analysis measuring light scattering from individual analytes in flow is a well-known high-throughput detection technique. Furthermore, flow cytometric analysis is compatible with other detection methods and analytical processes, such as fluorescent detection or selective sorting, because it enables in-line and non-invasive detection. From the above advantages, several scattering-based micro/nanoparticle detection systems in flow have been realized using microcapillaries and microchannels. For example, Norgaard et.al realized the quantification and sorting of metallic nanoparticle clusters using flow cytometric techniques (Simonsen et al. 2016). Yan et al. developed high sensitivity flow cytometer and applied it to bacteria, viruses, and extracellular vesicles (Tian et al. 2018; He et al. 2020). Yang et al. realized the counting of Au nanoparticles in flow by hydrodynamic focusing and total internal reflection microscopy (Liang et al. 2016).

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In general, flow cytometric analysis using microcapillaries and microchannels requires precise position alignment of analytes. Since the typical spot size of a focused laser beam (μm scale) is smaller than flow width and larger than analytes, only a part of nanoparticles passing through the spot can be detected, and scattering light intensity mainly depends on the positions and trajectories of analytes; each signal has little information about the size, shape, or refractive index of analytes. Hydrodynamic focusing has been widely used for focusing all analytes near the center of the laser spot. However, the focusing width is limited by the Brownian motion of analytes, and this effect is especially remarkable for small nanoparticles. For example, nanoparticles with 20 nm in diameter diffuse more than 1 μm within 50 ms by the Brownian motion. Furthermore, complicated device structure and precise fluidic control are inevitable for hydrodynamic focusing. Nanofluidic devices using 10^1–10^3 nm channels can solve the above problems, because positions and trajectories of nanoparticles are limited within a nanochannel. Since the width of a nanochannel is smaller than the laser spot, all nanoparticles pass through near the center of the laser spot by adjusting the position of the nanochannel. Due to the above advantage, nanofluidic devices have been applied to various nanoparticle analyses. For example, Manoharan et al. demonstrated scattering-based tracking of a single virus in nanofluidic optical fiber with 250 nm inner diameter (Faez et al. 2015). Novotny et al. used 500 nm wide nanochannels and realized scattering-based detection and size measurement of HIV viruses in flow (Mitra et al. 2012).

Although nanofluidic devices provide a suitable detection platform for nanoparticle analyses, the low detection performance for nanoparticles in flow still needs to be addressed. Different from measurements of fixed or suspended nanoparticles, flow cytometric analysis requires sensitive detection of nanoparticles passing through the detection region within microseconds to milliseconds. Furthermore, as described by Mie theory, scattering light intensity drops with the sixth power of the particle size; detection of nanoparticles smaller than 100 nm in diameter in flow by standard light scattering is challenging. Interferometric detection provides significantly better sensitivity for nanoparticles, because interferometric signal scales with the third power of the particle size (Ortega-Arroyo and Kukura 2012; Young and Kukura 2019). Interferometric scattering detection is classified into several types depending on the source of the reference light. The reflection of an illumination laser beam has been widely used as a reference light. For example, reflection-based interferometric light scattering microscopy has realized nanoparticle tracking in live cells, single protein imaging, interfacial dynamics measurement, and mass imaging (Young et al. 2018; Taylor et al. 2019; Chen et al. 2020). Such reflection-based interference is mainly used for nanoparticles immobilized on or near the surface which is the source of reflected light. For flow cytometric analyses, Novotny et al. realized interferometric detection of viruses flowing in nanochannels using a frequency-shifted reference beam (Mitra et al. 2010, 2012). However, a complicated optical system and precise phase adjustment are inevitable for such dual-beam interferometric detection, which might also lead to a large variance of measured signals. Thus, interferometric light scattering detection without a complicated optical system and precise optical adjustment is strongly desired for simple and sensitive nanoparticle detection in nanofluidic devices.

Here, we propose nanofluidic optical diffraction interferometry (NODI), interferometric light scattering detection of individual nanoparticles in flow using optical diffraction phenomena of a nanochannel. Optical diffraction by a single or multiple nanochannels has been previously used for refractive index measurement (Purr et al. 2019; Tsuyama and Mawatari 2020a, b, c), real-time DNA amplification monitoring (Yasui et al. 2016), and photothermal molecule/nanoparticle detection in nanochannels (Tsuyama and Mawatari 2020a, b, c; Tsuyama et al. 2021). In this study, we for the first time utilize optical diffraction by a single nanochannel as a reference light for interferometric detection. Since scattering light from nanoparticles in a nanochannel and diffracted light from the nanochannel itself follow almost the same optical path to the detector, interference of scattering light and diffracted light can be realized by simply irradiating a nanochannel with a single laser. Furthermore, all nanoparticles in the nanochannel pass through near the center of the laser spot, since the width of a nanochannel is smaller than the laser spot. These characteristics enable sensitive detection and signal-based classification of individual nanoparticles without complicated experimental systems and procedures, such as the introduction of another reference beam or position alignment of nanoparticles by hydrodynamic focusing. First, we demonstrate nanoparticle detection by the NODI, and the detection principle and characteristics are confirmed by several experiments. Then, the size measurement of individual nanoparticles is performed. Finally, by introducing dual-wavelength measurement, we differentiate plasmonic metal nanoparticles with different sizes and materials from their interferometric signals. The classification performance of the NODI is evaluated by a machine learning method.

2 Materials and methods

2.1 Detection principle of the NODI

Figure 1 shows the detection principle of the NODI. Since the typical spot size of a focused laser beam (μm scale) is

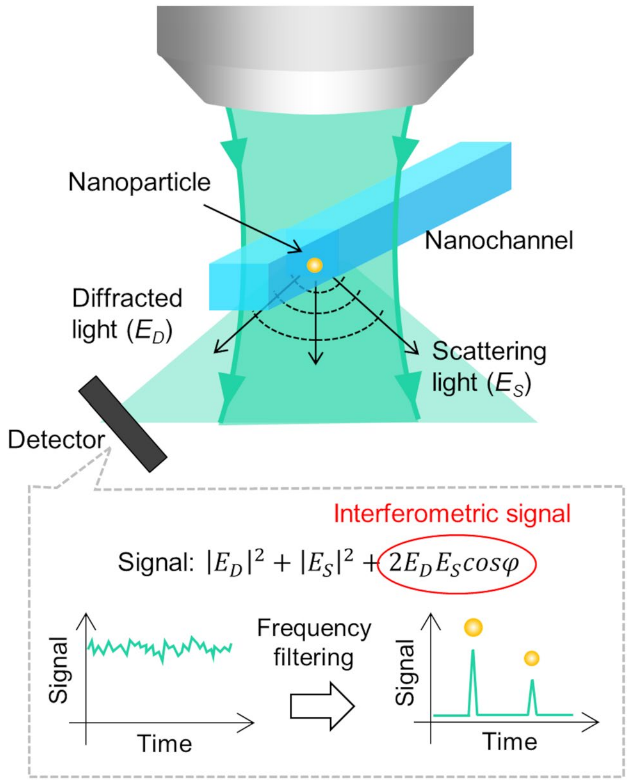

Fig. 1 Detection principle of the nanofluidic optical diffraction interferometry (NODI)

smaller than the width of a nanochannel, the focused laser beam is diffracted by the nanochannel. The NODI utilizes this nanochannel-specific optical diffraction effect as a reference light for the interferometric detection of individual nanoparticles. A nanoparticle in a nanochannel generates scattering light when flowing through the focused position of the laser beam, and the scattering light from the nanoparticle interferes with the diffracted light from the nanochannel. The total light intensity (I_t) at the detector is described as follows:

$$I_t = \left|E_D + E_S\right|^2 = \left|E_D\right|^2 + \left|E_S\right|^2 + 2E_D E_S \cos\varphi, \tag{1}$$

where E_D is the diffracted light field, E_S is the scattered light field, and φ is the relative phase difference. The first and second terms of the three components denote the contribution of pure diffracted light and pure scattering light, respectively. The third term denotes the interferometric component of the diffracted light and scattering light. Since pure scattering light scales with the sixth power of the particle size and is very weak (10^{-15} W) compared with pure diffracted light (10^{-9} W), it is extremely difficult to detect nanoparticles based on pure light scattering. In contrast, the third term scales with the third power of the particle size. Furthermore, the diffracted light field of nanochannels is usually more than 100 times stronger than the scattering light field of nanoparticles; the third term is on the order of 10^{-12} W. Thus, we can detect the interferometric component which has the information about the scattering properties of nanoparticles. For actual measurements, extracting the interferometric component from the first term (diffracted light component) is essential, because it is also more than 100 times stronger than the interferometric component. Although the diffracted light component is almost constant during a measurement time, the interferometric component is a sharp pulse signal, because nanoparticles pass through the detection position within a few milliseconds. Therefore, measuring high-frequency component (kHz order) by lock-in amplifier enables selective extraction of the interferometric component from the diffracted light component.

The advantages of the NODI over conventional scattering detection methods are as follows. First, the NODI provides simple yet highly sensitive nanoparticle detection by the interference of scattering light and diffracted light as mentioned above. Since nanochannel itself works as a source of reference light by its unique optical diffraction effect, simply irradiating a nanochannel with a single laser enables interferometric detection; neither another reference beam nor complicated optical adjustment is necessary. Such characteristics also enable multi-wavelength measurement or combination with other detection methods. Second, nanoparticles are confined in a nanochannel and pass near the center of the focused laser spot, ensuring that all nanoparticles in a nanochannel experience laser illumination. Therefore, each nanoparticle can be accurately measured without the position alignment of nanoparticles by hydrodynamic focusing. Although the relative phase difference of scattering light and diffracted light affect signal intensity as shown in the above equation, our detection method realized size measurement and classification of individual nanoparticles as shown later. Thus, highly sensitive detection and characterization of nanoparticles with a simple measurement system can be realized by making use of the characteristics of nanochannels.

2.2 Optical system

Figure 2 shows the optical system of the NODI. A 532 nm continuous-wave solid-state laser (Verdi-5M, Coherent Inc.) was introduced to a microscope after passing through a beam expander and focused on a nanochannel by an objective lens (20×, NA = 0.45). The calculated spot size was approximately 2 µm. The nanofluidic device was fixed on a 2-D stage (BIOS-235S, Sigma Koki Co., Ltd., Japan) which allowed precise adjustment of the channel position at 100 nm steps for each direction. Transmitted light, diffracted light by the nanochannel, and scattering light from nanoparticles were collected by another objective lens (NA = 0.9).

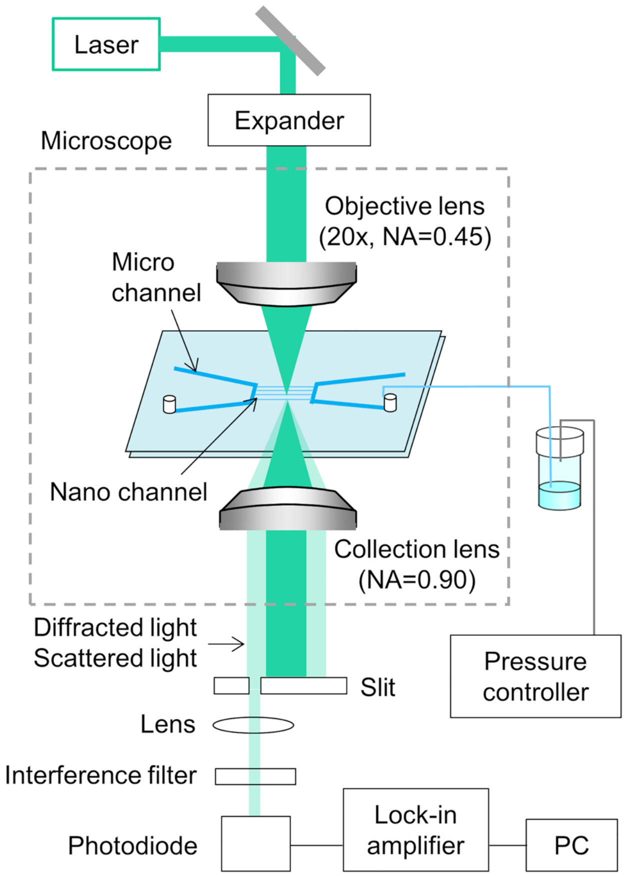

Fig. 2 Schematic optical system of the NODI

After cutting transmitted light by a slit with 1 mm width, signals were detected by a photodiode with a 400 µm aperture pinhole (ET-2030, Electro-Optics Technology Inc.). The high-frequency component was extracted by a lock-in amplifier (LI5660, NF Corporation, Japan) and recorded by a data logger for every 0.5 ms. The time constant of the lock-in amplifier was set to 1–2 ms. A 488 nm solid-sate laser (Sapphire488-500SV, Coherent Inc.) and a 660 nm solid-state laser (gem660, Laser Quantum Ltd., England) were used for measurements at different wavelengths.

2.3 Device fabrication and fluidic control system

The fabrication procedure of nanofluidic devices is shown in previous reports (Morikawa et al. 2020). Briefly, nanochannels were fabricated on a glass substrate (70 mm × 30 mm × 0.7 mm thickness, VIO-SIL, Shin-Etsu Quartz Co., Ltd., Tokyo, Japan) by electron-beam lithography and dry etching. Microchannels for sample introduction to nanochannels were fabricated on another glass substrate by photolithography and dry etching. After drilling inlet and outlet holes, these substrates were thermally bonded together at 110 °C for 3 h. The size of fabricated microchannels was 500 µm wide and 4.5 µm deep. The size of fabricated nanochannels was 800 and 1200 nm wide and 550 nm deep which were measured by scanning electron lithography (Regulus8230, Hitachi High-Technologies Corp., Japan) and stylus profiler (dektakXT, Bruker Corp.).

Sample injection and replacement were performed by applying pressure to inlets of nanofluidic devices. Details of the pressure-driven fluidic control system are described elsewhere (Ishibashi et al. 2012). Briefly, sample solutions stored in vials were introduced to nanochannels through capillaries with 0.26 mm inner diameter (Institute of Microchemical Technology Co., Ltd, Japan) and microchannels by applying pressure using a pressure controller (MFCS-EZ, Fluigent Inc., France). The outlet of the microchannel was opened to avoid the clogging of nanochannels by nanoparticle agglomerates. The flow rate was calculated from the replacement time of the sample solution in nanochannels which was measured by our previously developed detection method (Tsuyama and Mawatari 2020a, b, c). The calculated flow rate in the nanochannel for 100 kPa pressure was 0.2 mm/sec, which corresponds to 5.4 pL/min. In this flow rate, individual nanoparticles pass through the detection region within 10 ms.

2.4 Reagents and samples

Gold and silver nanoparticle solutions were purchased from Sigma-Aldrich Co. LLC. 20 and 30 nm gold nanoparticle solutions were diluted to 100 pM (6.0×10^{10} particles/mL). Other nanoparticle solutions were used without dilution. The concentration of these solutions was ~ 10^{10} particles/mL. The expectation value of the number of nanoparticles in the laser focus calculated from the focused laser spot diameter and cross-sectional area of the nanochannel was 5.7×10^{-2} at 100 pM sample concentration. Considering the Poisson distribution, the probability of detecting two or more particles for the probability of detecting a single particle is less than 3%. All solutions were filtered with a 0.22 µm syringe filter to avoid the clogging of nanochannels by nanoparticle agglomerates.

3 Results and discussion

3.1 Verification of nanoparticle detection

The detection principle of the NODI was verified by the following experiments. First, we checked signals for three different experimental conditions shown in Fig. 3. As a test sample, a 30 nm gold nanoparticle aqueous solution was introduced to channels by applying pressure at 100 kPa. The detection frequency was set to 3 kHz. We took images of the laser spot after passing through the collection lens for

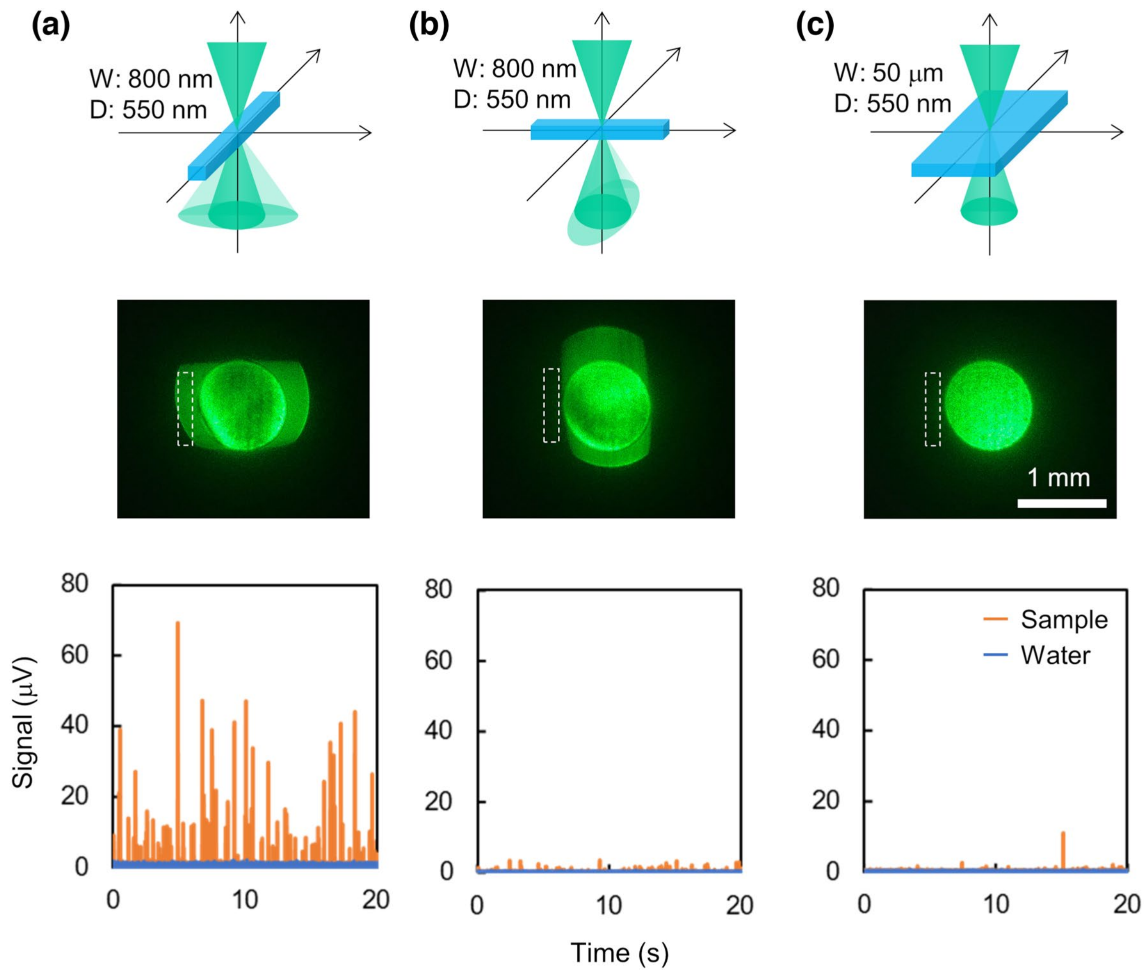

Fig. 3 Summary of the images of laser spots for blank channels and measured signals for different geometries. The dotted square area shows the corresponding detection position by a slit. **a** 800 nm wide channel and measuring the diffracted light region. **b** 800 nm wide channel and measuring the out of the diffracted light region. **c** 50 μm wide channel and measuring the region perpendicular to the length of the channel

blank channels filled with air, and the dotted square shows the corresponding detection position by a slit for the actual experiment. Figure 3a shows the results for 800 nm wide nanochannels when measuring the diffracted light region. We observed many pulse signals derived from nanoparticles. In contrast, as shown in Fig. 3b, very weak pulse signals were observed when measuring the out of the diffracted light region. Figure 3c shows the results for 50 μm wide channels when measuring the region perpendicular to the length of the channel. Since the width of the microchannel is larger than the spot size of the focused laser beam, no diffracted light was observed. Obtained results were almost the same as those for nanochannels when measuring outside the diffracted light region. Thus, observed pulse signals in Fig. 3a were not pure light scattering from nanoparticles but related to the diffracted light. Note that the light scattering by the roughness of the wall surfaces is not considered in this study, since our previous work shows the diffraction phenomena is almost consistent with the theory, suggesting the scattering by the roughness is negligible compared to the diffracted light by the nanochannel itself (Tsuyama and Mawatari 2020a, b, c).

Next, we investigated the dependence of pulse signals on the detection position by scanning the slit from the center region of the transmitted light to the diffracted light region. The radius of the beam spot was adjusted to 5 mm at the slit position, and the slit width was set to 1 mm. In this study, a 40 nm gold nanoparticle aqueous solution was introduced to channels by applying pressure at 300 kPa. Figure 4 shows the pulse counts versus slit position from the center of the beam spot. We counted the number of pulse signals which is higher than the background signal $+10\sigma$ with a duration time of more than 2.5 ms. The interval of the counting was set to more than 100 ms to avoid multiple counting of the same nanoparticles. Many pulse signals were observed at the diffracted light region, while the number was significantly

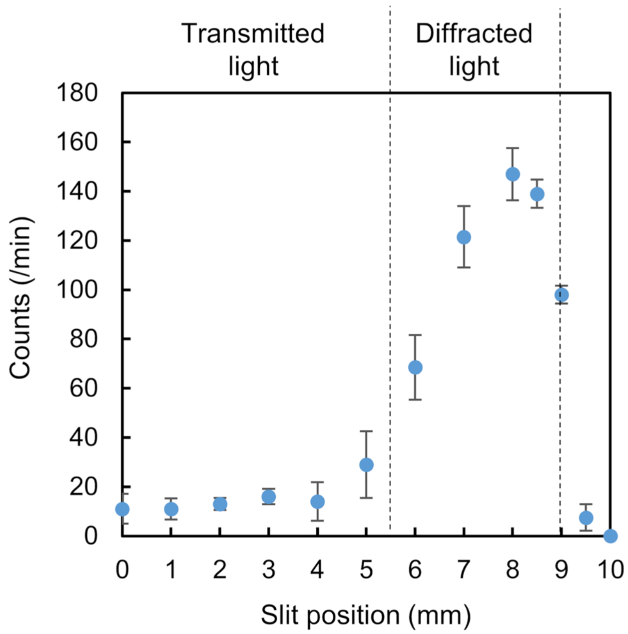

Fig. 4 Pulse counts versus slit position from the center of the beam spot. Dashed lines show diffracted light region

decreased at the transmitted light region. The average pulse height increased depending on the diffracted light intensity (Figure S1). These results are consistent with the proposed detection principle.

Furthermore, we confirmed the signal generation mechanism by measuring signals for different nanoparticles and wavelengths. Since plasmonic nanoparticles have a distinctive absorption and scattering spectrum for a visible light region, a comparison of experimental results with theoretical calculations reveals whether the signal generation mechanism is based on the absorption, scattering, or interferometric scattering. To eliminate the effect of different optical components for each wavelength, we measured signals of 40-nm silver and gold nanoparticles for 488 nm, 532 nm, and 660 nm wavelengths and compared the ratio of the signal intensity (silver/gold). Theoretical values were calculated by Mie theory for homogeneous spheres (Jain et al. 2006; Zhang 2011). The physical properties of nanoparticles are according to the literature values (Johnson and Christy 1972). Theoretical values for interferometric scattering were calculated as a square root of those for pure scattering. Details of theoretically calculated values are shown in supporting information. Figure 5 summarizes calculated theoretical values and experimental results. Since gold nanoparticles show higher absorbance efficiency than silver nanoparticles around measured wavelength, the calculated signal ratio for absorbance is smaller than 1. In contrast, silver nanoparticles show higher scattering efficiency than gold nanoparticles for 488 nm wavelength, and whether pure

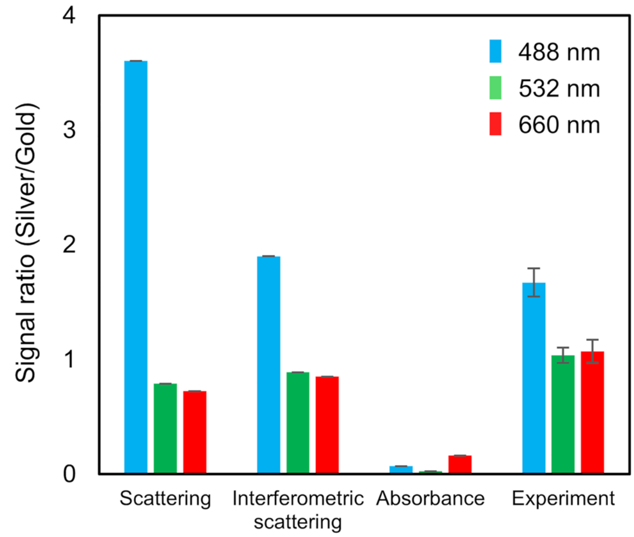

Fig. 5 Signal ratio of silver and gold nanoparticles for different wavelengths. Theoretical values were calculated based on Mie theory

scattering or interferometric scattering can be decided by the signal ratio. The measured ratios of the signal intensity for three wavelengths were well consistent with theoretical ones for interferometric scattering. Thus, the proposed detection principle of the NODI was verified.

3.2 Effects of detection frequency

Our detection system extracts an interferometric component from diffracted light by frequency selective detection using a lock-in amplifier. Therefore, signal intensity and the detectable number of counts are highly dependent on the detection frequency by the lock-in amplifier. Figure 6 shows

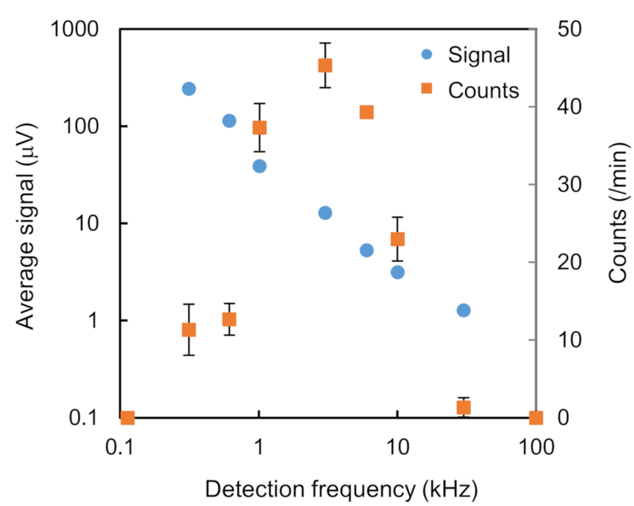

Fig. 6 Average signal intensity and pulse counts per minute versus detection frequency

the average pulse height and counts for different detection frequencies. The experiment was repeated three times. The average pulse height was inversely proportional to the detection frequency, and the number of counts decreased in the high-frequency region. The number of counts also decreased in the low-frequency region, because background and noise derived from the diffracted light become dominant, and the optimum detection frequency was around 1–6 kHz. Although we extracted an interferometric component at a specific detection frequency, bandpass or high-pass frequency filtering will lead to further sensitivity improvement. Note that optimum detection frequency also depends on the flow rate, optical components, and external noise.

3.3 Size measurement of nanoparticles

In the detection system of the NODI, nanoparticles are confined in a nanochannel and pass near the center of the focused laser spot, ensuring almost the same laser illumination for all nanoparticles without hydrodynamic focusing. To evaluate the detection performance of the NODI, we checked the relationship between signal intensity and the size of nanoparticles. Figure 7 shows a logarithmic plot of the average signal intensity for gold nanoparticles of different sizes. 500 counts of pulse signals were measured for each particle size. Outliers were eliminated based on the interquartile range (IQR). In this study, pulse signals having a value more than 1.5 IQR below Q1 or more than 1.5 IQR above Q3 were eliminated as outliers. Measured average signal scaled with the 2.3th power of the particle diameter, indicating signals were derived from not the pure scattering (scaling with the sixth power of the particle size) but the interference (scaling with the third power of the particle size). The measured average signal-to-noise ratio was 12 for 20 nm gold nanoparticles and 33 for 30 nm gold nanoparticles. Although pulse signals for 20 nm gold nanoparticles were observed, not all nanoparticles might be detected due to the signal-to-noise ratio. Therefore, we evaluated the signal intensity distribution for 30 nm gold nanoparticles. Figure S2 shows the intensity distribution of 2000 pulse signals for 30 nm gold nanoparticles. The coefficient variation (CV) was estimated by Gaussian fitting. The calculated CV was 20.3%, corresponding to ± 6.1 nm deviation. Although the unimodal distribution of particle size was confirmed, the CV was larger than those reported by the manufacturer ($<12\%$). The difference in particle size dependency and CV for predicted values is discussed later.

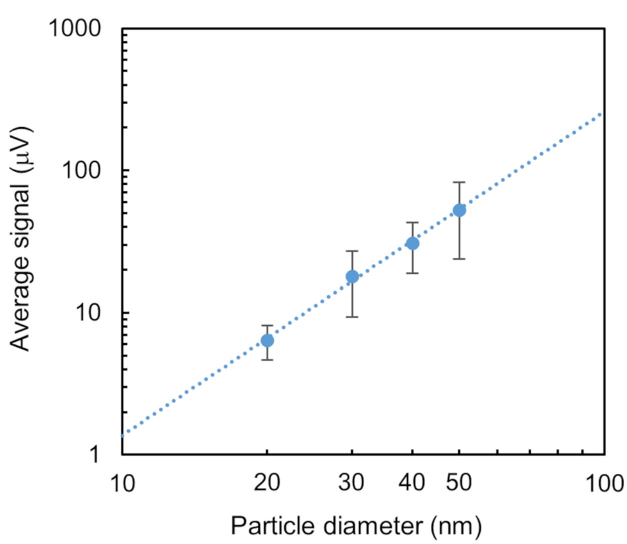

Fig. 7 Average signal intensity versus particle diameter of gold nanoparticles

3.4 Nanoparticle classification by dual-wavelength measurement

Finally, we discriminated plasmonic metal nanoparticles of different sizes and materials by dual-wavelength NODI. Gold and silver nanoparticle solutions with 40 and 60 nm diameters were used as a test sample. Two laser lines with a wavelength of 488 nm and 532 nm were used to detect and classify these four types of nanoparticles. Gold nanoparticles have a higher scattering efficiency at 532 nm than 488 nm, whereas silver nanoparticles have a higher scattering efficiency at 488 nm than 532 nm. The optical system for dual-wavelength measurement is shown in Fig. S3. Two lasers were super-positioned and focused on a nanochannel, and each diffracted light was measured by a photodiode and lock-in amplifier. Figure S4 shows the example of signals for dual-wavelength measurement. Pulse signals for 488 nm and 532 nm wavelengths were measured without signal crosstalk. We extracted pulse signals for 488 nm wavelength which is higher than the background signal $+10\sigma$ with a duration time of more than 2.5 ms, and each signal was linked with corresponding signals for 532 nm wavelength by searching for a maximum value within the pulse width. Figure 8 shows a scatter plot of the signal intensity for dual-wavelength measurement. Four types of nanoparticles were classified with a little overlap. Gold nanoparticles show higher scattering intensity for 532 nm than 488 nm, whereas silver nanoparticles show higher scattering intensity for 488 nm than 532 nm, which is well consistent with theoretical calculations.

Furthermore, we evaluated classification performance by applying a machine learning method. Support vector machine (SVM) algorithm was used for the classification of four types of nanoparticles from their pulse heights and widths. Data processing and SVM classification were implemented using Python version 3.7.6 and the scikit-learn library (Python Software Foundation, Wilmington,

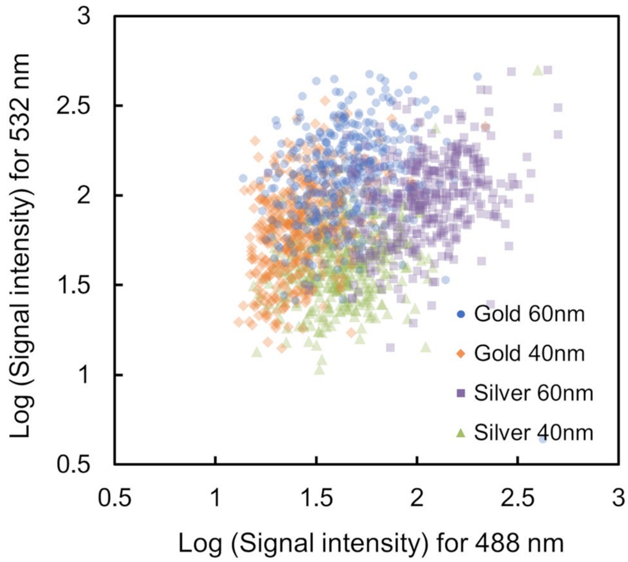

Fig. 8 Scatter plot of logarithmic signals for dual-wavelength measurement of gold and silver nanoparticles

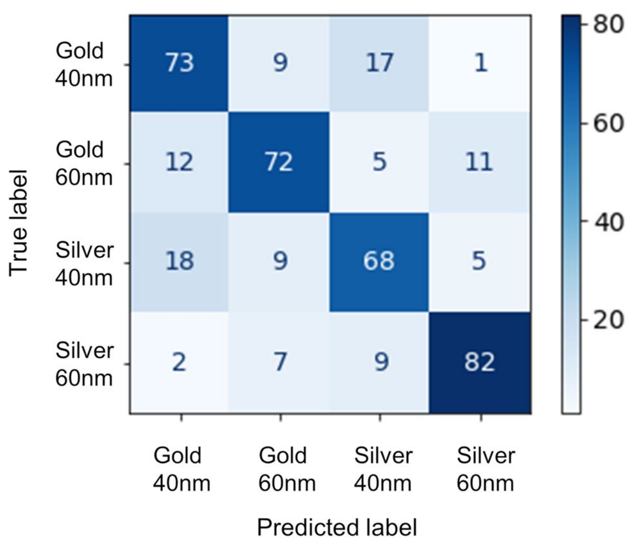

Fig. 9 Confusion matrix for the support vector machine (SVM) classification of four types of nanoparticles. 400 data sets for each nanoparticle were used with a 75:25 train/test split

DE, USA). In total, 1600 data sets were used for training, validation, and testing the classifier. The detailed procedure is shown in supporting information. Radial basis function kernel was used for classification, and the cost parameter and kernel parameter were set to 1.0 and 0.1, respectively. The average accuracy score for tenfold cross-validations was 71.9 ± 4.0%. Figure 9 shows the confusion matrix for the classification of four types of nanoparticles. Although nanoparticles with similar size or the same material were misjudged to some extent, four types of nanoparticles were discriminated based on pulse signals for 488 nm and 532 nm wavelengths. We also confirmed the accuracy score for the two types of nanoparticles as more than 80% (Figure S5). Thus, metal nanoparticles smaller than 60 nm diameter was successfully discriminated by the NODI and SVM classification. In this study, only the height and width of pulse signals for each wavelength were used as feature parameters for classification. Additional feature parameters such as area, onset angle, and bluntness of pulse signals might improve the classification performance.

The above results show that the NODI enables the size measurement and classification of nanoparticles based on their signals to some extent. As shown in the experimental section, interferometric signals are dependent on the relative phase difference. In conventional interferometric scattering detection methods using reflected light or another beam, fixation of nanoparticles or precise optical adjustment is essential to minimize the effect of relative phase difference and get information of individual nanoparticles from their signals. In contrast, the NODI realized the detection and characterization of nanoparticles in flow without any phase adjustment. The possible reason is as follows. In the detection system of the NODI, the phase of scattering light will change drastically while passing through the detection region due to the movement of nanoparticles in the flow direction and the Brownian motion. Therefore, obtained interferometric scattering signal also fluctuates at a value below the maximum signal intensity determined by the property of each nanoparticle. By extracting a specific frequency component at a millisecond time constant, averaged pulse signals can be obtained from the fluctuated interferometric scattering signals. Since these pulse signals still reflect the scattering property of corresponding nanoparticles, size estimation and classification of individual nanoparticles can be realized. The difference in particle size dependency and CV relative to predicted values might be attributed to such measurement and signal processing. Note that other factors such as the fluctuation of diffracted light also contribute to the deviation of the measured signal. Although the above hypothesis is still under investigation, our experimental results show that simple and sensitive scattering-based nanoparticle detection and classification are feasible using nanofluidic optical diffraction.

4 Conclusions

In this study, we developed nanofluidic optical diffraction interferometry (NODI) and realized the detection and classification of individual nanoparticles smaller than 100 nm in size flowing in a nanochannel. Our detection method does not need a complicated experimental system, such as the introduction of another reference beam for interference and position alignment of nanoparticles by hydrodynamic focusing. Therefore, other detection techniques such as absorption and fluorescence measurement can be easily combined. In addition, single nanoparticle sorting based on measured signals will be realized by an integrated micro/nanofluidic system. In this study, we demonstrated dual-wavelength measurement and classification of plasmonic metal nanoparticles by machine learning method. These results suggest that more accurate label-free characterization of plasmonic nanoparticles can be realized by measuring the scattering spectrum using multiple wavelength illumination. Our method can also be applied to other common nanoparticles such as silica particles and extracellular vesicles that do not have a characteristic scattering spectrum, since their scattering signal intensity gives information about sizes and refractive indices. As shown in the theoretical formula in the experimental section, a signal-to-background ratio of the NODI is related to the diffracted light intensity which depends on the channel size, refractive index difference between substrate and solvent, and wavelength of the laser. Therefore, the detection sensitivity can be further improved by optimizing such parameters. One of the bottlenecks of using nanochannels is the low flow rate (pL/min), which leads to low detection throughput. In this study, the detection throughput was on the order of 10^1–10^2 particles/min for the measured concentration range. Increasing flow rate by applying high pressure or preconcentration of sample solution will improve the throughput. Although there is still room for various improvements, we believe that our detection method will become a powerful and versatile tool for nanoparticle analysis using nanofluidic devices.

Appendix

The Supporting Information is available free of charge at (URL).

Average signal intensity versus diffracted light intensity, histogram of the signal intensity for 30 nm gold nanoparticles, experimental setup and example of measured signals for the dual-wavelength measurement system, accuracy score for classification of two types of nanoparticles by machine learning method, theoretically calculated values of scattering, absorption, and interferometric scattering for 40-nm gold and silver nanoparticle, and procedure of classification of four types of nanoparticles using machine learning.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10404-022-02562-y.

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Declarations

Conflict of interest The authors declare no competing financial interest.

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